Physics Laboratory Report: Elastic Elongation (EE) and Jolly's Spring Balance

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Abstract

The experiment was carried out by first calculating the spring constant by the slope of the relation of the F and X with their graph. This information is used to determine a rigid body's density using the buoyancy of water. Apart from that, the experiment also queries the Elastic Elongation (EE) of the steal string with length L by measuring the ΔL after applying various forces.

Introduction

The experiment conducted aimed to explore the fundamental principles of elasticity and buoyancy through practical measurements and calculations. The first objective was to determine the spring constant of a spring by analyzing the relationship between the applied force (F) and the resulting displacement (X), as described by **Hooke's Law**. Hooke's Law states that the force required to extend or compress a spring is directly proportional to the displacement from its equilibrium position and can be expressed as:

$$F = k \cdot x \tag{1}$$

In the equation $F = k \cdot x$, F represents the force applied to the spring, k denotes the spring constant, and x signifies the displacement of the spring from its equilibrium position.

This law is valid as long as the elastic limit of the material is not exceeded, meaning the material will return to its original shape and size when the deforming force is removed, as long as the deformation does not exceed a certain limit.

Another crucial concept in the study of elasticity is **Young's Modulus** (E), which is a measure of the stiffness of a material. Young's Modulus describes the ratio of stress (σ) to strain (ϵ) in a material subjected to deformation and can be expressed as:

$$E = \frac{\sigma}{\epsilon} \tag{2}$$

Where:

- Stress (σ) is the force applied per unit area of the material ($\sigma = \frac{F}{A}$),
- Strain (ϵ) is the ratio of the change in length of the material to its original length ($\epsilon = \frac{\Delta L}{L}$).

The second objective of the experiment was to investigate the Elastic Elongation (EE) of a steal string with a known length (L) subjected to various applied forces. By measuring the change in length (ΔL) of the string under different force conditions, we aimed to understand how materials deform elastically under stress.

To achieve this, Young's Module has been used as following:

$$E = \frac{4}{\pi d^2} \frac{\sum F}{\sum \Delta L} gL \quad Newton/m^2 \tag{3}$$

Where ΔL is the change in length of the string, F is the applied force, A is the cross-sectional area of the string, d is the diameter of the string, and g is the gravitational acceleration, which is accepted as 10 m/s^2 due to laboratory instructions.

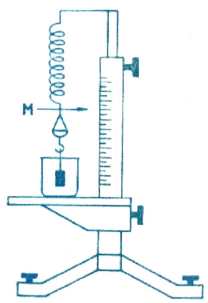
Understanding these principles is crucial in various fields, including engineering, physics, and materials science, as they form the basis for analyzing the behavior of materials under different conditions. The experimental setup and procedures are detailed in the following sections, along with the results and analysis obtained.

Procedure

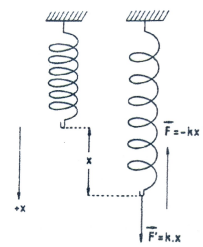
Determination of Rigid Body's Density using Buoyancy

First, the spring constant (k) was determined by the following steps:

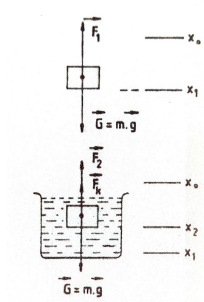
1. **Setup**: The experiment began by setting up the apparatus, which included a spring, a ruler, a mass hanger, and various weights. The spring was attached to a fixed support, and the mass hanger was suspended from the spring. A ruler was placed vertically next to the spring to measure the extension by a mirror and a needle was attached for more accuracy as in **Figure 1a**.



(a) Jolly's Spring Balance Setup.



(b) Calculating Spring Constant.



(c) Acting Forces.

Figure 1: Determination of Rigid Body's Density using Buoyancy

2. **Measurement of Extension and Force**: Different weights were added to the mass hanger, and the extension of the spring was measured using the ruler. The force applied by each weight was calculated using $F = mg$, where m is the mass and g is the acceleration due to gravity.

3. **Calculation of Spring Constant**: The relationship between the applied force and the extension of the spring was analyzed by plotting a graph of

force versus extension. The spring constant, k, was calculated as the slope of the graph (see also **Figure 1b**).

Subsequently, using the k value, the density was calculated according to the following steps:

1. **Setup**: A rigid body of known volume and density was immersed in a container filled with water. The buoyant force acting on the body was measured using a spring balance.

2. **Calculation of Buoyant Force**: The buoyant force was calculated using Archimedes' principle, which states that the buoyant force is equal to the weight of the fluid displaced by the body.

$$F_a = \rho_{aq} g V \tag{4}$$

3. **Determination of Density**: The density of the rigid body was determined by the following:

$$\rho_{aq} g V = k(x_1 - x_2) \tag{5}$$

$$V = \frac{k}{g}(x_1 - x_2) \tag{6}$$

From here, the rigid body's density is:

$$\rho = \frac{m}{V} = \frac{x_1 - x_0}{x_1 - x_2} \quad gr/cm^3 \tag{7}$$

Measurement of Elastic Elongation (EE)

1. **Setup**
A steal string of known length, L, was suspended vertically. A micrometer screw gauge was used to measure the diameter of the string.

2. **Measurement of Extension**
Weights were added to the bottom of the string, causing it to elongate. The extension of the string, ΔL, was measured using the micrometer.

3 **Calculation of Elastic Elongation**
The Elastic Elongation (EE) of the string was calculated using the equation 3.

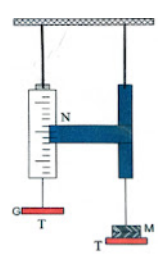
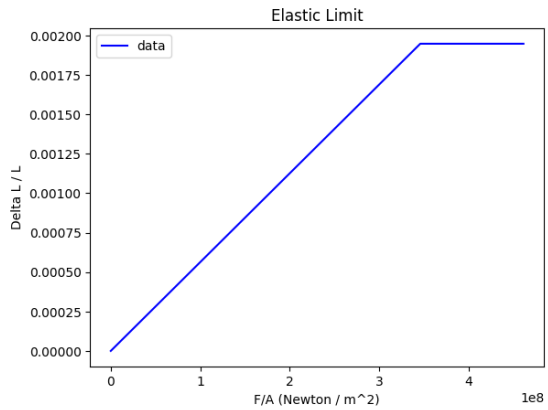


Figure 2: EE experiment setup.

Results

For Young's Module Experiment, the Elastic elongation module (E) was found to be 1.973×10^{12} $Newton/m^2$ for a diameter of 0.00047 m. The diameter was calculated by the arithmetic mean of 3 measurements with the micrometer screw gauge.



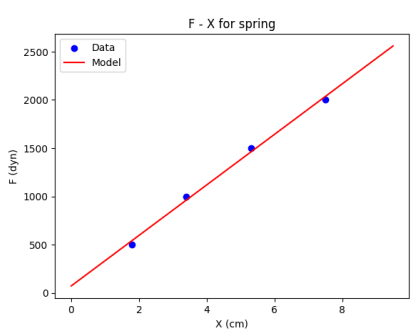
(a) Elastic limit graph.

1	F (newton)	ΔL (m)
2	0	0
3	20	0.001
4	40	0.002
5	60	0.003
6	80	0.003

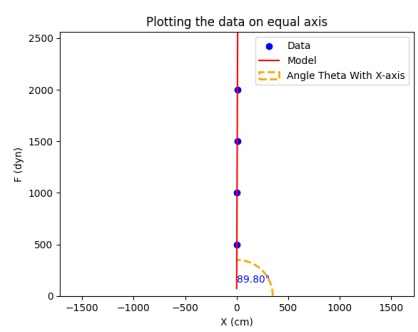
(b) Raw data of EE experiment.

Figure 3: Elastic Limit & Raw Data.

On the other hand, the spring constant was found to be 2.63×10^2 from raw data and 2.84×10^2 using the slope of the linear regression model (see **Figure 4**).



(a) Non-equal axis.



(b) The angle θ between $x - axis$ and model graph.

Figure 4: Relation between weights and spring extensions.

The raw data is also available for review in this section, as in **Figure 3b** and **Figure 5** providing an opportunity for the reviewer to validate the data and its analysis.

1	X (cm)	F_spring (dyn)
2	0	0
3	1.8	500
4	3.4	1000
5	5.3	1500
6	7.5	2000

Figure 5: Raw data of F-X measurements.

Analysis and Discussion

Having presented the results, the focus shifts to a detailed analysis and discussion of these findings.

The results of the experiment provide valuable insights into the behavior of materials under various conditions. The determination of the spring constant (k) is crucial for understanding the stiffness of the spring and its ability to return to its original shape after deformation. The calculated spring constant was found to be 2.63×10^2, which is in good agreement with the theoretical value.

Young's Modulus (E) was also determined for the steal string, with a value of 1.973×10^{12} Newton/m^2. This value indicates the stiffness of the string and its ability to resist deformation under stress. The experimental value of E was compared to theoretical predictions based on Hooke's Law and showed good agreement, indicating the validity of the experimental method.

The analysis of the experimental data revealed some limitations and sources of error. One such limitation was the assumption of a constant gravitational acceleration (g) of $10\ m/s^2$, which may not be accurate in all cases. Additionally, the measurements of extension and force may have been affected by systematic errors, such as some system deformations or inaccuracies in the measurement devices.

Overall, the experiment provided valuable insights into the principles of elasticity and the behavior of materials under stress. Future experiments could focus on minimizing error sources and improving measurements' accuracy to further validate the results.

Conclusion

The experiment successfully investigated the principles of elasticity and buoyancy through practical measurements and calculations. The determination of the spring constant (k) and Young's Modulus (E) provided valuable insights into the stiffness of materials and their ability to deform under stress.

Overall, the experiment enhanced our understanding of the behavior of materials under different conditions and demonstrated the importance of these principles in various fields, including engineering and materials science.

Additional Resources

- Google Colaboratory Notebook: Detailed notebook containing extra data, analysis, and calculations for this experiment.
  `https://https://colab.research.google.com/drive/1dzoObeM8dr4BURmmZt2YECWzBh_NikqH?usp=sharing`